

TOTEM



THE INTELLIGENCE EDITION

Usecase Bible

A verifiable operating fabric
for the autonomous economy.

People. Applications. Agents. Machines. Institutions.

Rewritten around @totemsdk/intelligence and @totemsdk/qvac, with a
complete review of the original application universe.

36

APPLICATION DOMAINS

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INTELLIGENCE DOMAINS

READ / NAVIGATE

The whole universe, in one place

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Each domain includes the original scope, the intelligence contribution, a composition pattern and its boundary.

STRATEGIC ASSESSMENT

Intelligence expands the fabric

The largest change is a shared interpretation and proposal layer across all 36 domains.

Totem Edge is a verifiable operating fabric for the autonomous economy, connecting people, applications, AI agents, machines and institutions through identity, evidence, intelligence, policy, governance, physical action and programmable settlement.

The two additions make AI an explicit SDK integration surface. intelligence defines provider-neutral contracts; qvac adapts the QVAC runtime to those contracts. Applications can combine understanding, retrieval, speech, generation and embodied-task inference with Totem's existing rights and economic systems. [S1-S3]

THE OPERATING PRINCIPLE

AI proposes. Totem authorises. An inference result can inform a decision; it does not itself grant a right, sign a transaction, certify a fact or make a machine action safe.

What changes in the application universe

Every existing vertical gains an assistant	Passports, finance, logistics, servicing, institutions and communities can expose evidence through source-linked, multilingual and potentially local interfaces.
Intelligence becomes a service to buy and sell	Inference, retrieval, transcription, generation and specialised model services can join agent budgets, service entitlements and settlement relationships.
Physical AI gains an explicit model seam	VLA and world interfaces support task-proposal and generative-preview designs, with separate engineering validation and bounded control.

Facts. Rights. Decisions. Actions. Value. These remain the five business flows. Intelligence helps people and agents interpret facts and propose decisions; the trust, policy, authority and settlement boundaries continue to matter.

Assessment based on source interfaces at commit e97b2c1c462d. Product concepts are compositions, not claimed production verticals.

ARCHITECTURE

Five systems. One operating loop.

Sense → Prove → Decide → Act → Settle remains the organising architecture.

SYSTEM	RESPONSIBILITY
Edge	Observe and interface with the world. Protocol readings, device identity and local services supply inputs; compatible hosts may expose an intelligence port.
Trust	Establish provenance, issuer relationships and verifiable claims. Keep source evidence distinguishable from generated interpretations.
Policy	Evaluate permitted behaviour, budgets, credentials and governance decisions. Bind an authenticated principal to a validated proposed operation.
Action	Translate a valid mandate into a bounded digital or physical effect. Engineering interlocks and control systems enforce the operating envelope.
Settlement	Record and transfer value or rights under the agreed economic state. Omnia, wallet and other settlement surfaces serve the relevant application.

Where intelligence fits

An application may invoke intelligence before or within decision-making: read a document, retrieve a rule, transcribe an observation, generate a forecast or propose a task. It should preserve the source, model and processing context so later policy and human review can assess the proposal. [S1-S4]

TWO DISTINCT CLAIMS

“This sensor signed this reading” is a provenance claim. “The model thinks a repair is needed” is an interpretation. A signed record of the latter can prove who recorded it; it still does not prove the diagnosis or the correctness of the computation.

Architecture and original application scope [S7]. Intelligence integration and implementation boundaries [S1-S5].

DELIVERY

Choose the host for the workload

The human signing surface and the inference runtime need not live in the same application or on the same device.

SURFACE	ROLE AND DEPLOYMENT IMPLICATION
Web application	@totemsdk/connect gives a dApp its wallet and application integration surface through the Totem Browser Extension. A web UI can request an approved service from an appropriate host.
Browser Extension	The self-custodial signing surface for human approval. Do not infer that adding qvac to the SDK bundles native model execution into the extension.
PWA wallet	An installable and cross-origin embeddable wallet in the supplied delivery model. Inference placement and browser compatibility must be established separately.
Axia services	Hosted chain/RPC access, broadcast, live portfolio infrastructure and WOTS lease coordination in the baseline. These roles do not establish a hosted QVAC service.
Backend / agents / MCP	Services can host the intelligence integration where supported, authenticate requests, enforce policy and connect to signing or settlement through separate authorised surfaces.
Edge / Pear hosts	A compatible workstation, gateway or device can host local inference after runtime, plugins and models are provisioned. Physical endpoints can continue using smaller protocol and proof components.

LOCAL IS A DEPLOYMENT PROPERTY

qvac accepts an injected SDK, a loader or a lazy runtime resolution path. Native binaries, model weights, plugins, memory, accelerators and offline assets still have to exist on the chosen host. A disconnected inference session and an offline payment/access workflow are separate designs. [S2]

Baseline delivery map [S7]; qvac runtime loading and Edge port implementation [S1, S2, S4].

PACKAGE ASSESSMENT

Contracts and execution are separate

Both Totem package manifests are version 0.1.0 in the inspected source. This review does not establish registry publication or production readiness. [S6]

LAYER	VERIFIED CONTRIBUTION / APPLICATION RESPONSIBILITY
intelligence	Provider-neutral types for domains, capabilities, operations, outcomes, streams, usage, receipts and errors; a provider interface and Edge-compatible port factory. Zero runtime dependencies.
qvac	A concrete QVAC provider, typed domain adapters, an Edge port and a raw SDK escape hatch. Its vendored type surface targets @qvac/sdk 0.19.0; the heavy runtime is resolved separately.
What they bring together	A reusable invocation seam across language, retrieval, speech, documents, media, embodied inference and model operations. Hosts can correlate requests with proposal/run context and build governed application flows.
What remains to compose	Identity validation, operation/parameter allowlists, policy enforcement, budget reservation, tariff conversion, receipt attestation, service verification and authorised downstream execution.

The capability count, correctly interpreted

The canonical QVAC snapshot lists 16 domains and 54 wrapped operations. This counts the adapter surface, including model and system administration. It does not mean 54 independent AI models, 16 loaded capabilities, or 54 production products. [S3]

PROVIDER-NEUTRAL DOES NOT MEAN EVERY PROVIDER IS IMPLEMENTED

The intelligence contract can accommodate other local or remote providers. This assessment verified the QVAC adapter; it does not claim bundled adapters for every named AI provider or automatic routing between them.

Package definitions [S1, S2, S6]. Canonical operation inventory [S3].

The 16-domain intelligence surface

Exact operation names from the QVAC 0.19.0 adapter snapshot; examples are composable application uses. [S3]

llm 03 OPS	completion, batchCompletion, finetune Text completion, batch work and fine-tuning interfaces; assistants, drafting and specialised model services.
embed 01 OPS	embed Vector representations for semantic indexing and retrieval workflows.
rag 09 OPS	ragChunk, ragIngest, ragSearch, ragSaveEmbeddings, ragDeleteEmbeddings, ragReindex, ragListWorkspaces, ragCloseWorkspace, ragDeleteWorkspace Chunk, ingest, search and manage embedding workspaces for source-linked knowledge services.
asr 04 OPS	transcribe, transcribeStream, bciTranscribe, bciTranscribeStream Speech transcription and streaming, plus BCI transcription wrappers; field notes and accessible interfaces.
translate 01 OPS	translate Language translation for support, documents and cross-border collaboration.
tts 02 OPS	textToSpeech, textToSpeechStream Speech synthesis and streaming for spoken guidance and accessibility.
diffusion 02 OPS	diffusion, upscale Image generation and upscaling for creative and visual-service workflows.
ocr 01 OPS	ocr Text extraction from documents and images; intake for passports, invoices and evidence packs.

A callable SDK function is not proof of a loaded model, compatible hardware, inference quality or an authorised operation.

The 16-domain intelligence surface

Exact operation names from the QVAC 0.19.0 adapter snapshot; examples are composable application uses. [S3]

classify 01 OPS	<code>classify</code> Classification interface for suitable models and supported inputs; routing and triage.
audiogen 01 OPS	<code>audioGen</code> Audio generation interface for licensed creative services.
video 01 OPS	<code>video</code> Video generation interface for creative and interactive applications.
vla 05 OPS	<code>vla, vlaHparams, vlaSetEmbodiment, vlaPreprocessImage, vlaPadState</code> Vision-language-action inference and embodiment/input helpers; proposed embodied tasks.
world 02 OPS	<code>worldCreateScene, worldStep</code> Scene creation and stepping for generative previews; not validated physical simulation.
models 13 OPS	<code>loadModel, unloadModel, getModelInfo, getLoadedModelInfo, deleteCache, downloadAsset, assessModelFit, modelRegistryList, modelRegistrySearch, modelRegistryGetModel, suspend, resume, state</code> Model and registry lifecycle interfaces for a provisioned local-model service.
system 06 OPS	<code>heartbeat, getSystemResources, loggingStream, subscribeServerLogs, cancel, close</code> Health, resources, logs, cancellation and close interfaces for host operations.
plugins 02 OPS	<code>invokePlugin, invokePluginStream</code> Plugin invocation and streaming; each plugin adds its own runtime, permissions and behaviour.

A callable SDK function is not proof of a loaded model, compatible hardware, inference quality or an authorised operation.

REFERENCE OPERATING MODEL

From inference to an authorised effect

The host must make the policy boundary real; the compute adapter does not enforce the complete chain for the application.

1 / Establish scope	Authenticate the principal; identify the proposal/run; determine consent, data access, allowed domain, canonical operation, model and plugin. Do not trust self-declared context.
2 / Approve compute	Validate the request and reserve an appropriate budget. Select a provisioned runtime and check the actual operation, model and resource requirements.
3 / Run and account	Invoke or stream. Wait for final runtime statistics where available; record sources and model version. Handle missing usage, partial output and best-effort cancellation.
4 / Validate the proposal	Treat retrieved text and model output as untrusted input. Parse an explicit schema; check recipient, amount, physical limits and evidence. Escalate where the mandate requires it.
5 / Authorise the effect	Apply policy and authority to the proposed digital or physical action. A governed signing surface approves only the bounded mandate; the controller enforces its envelope.
6 / Prove and settle	Record action/outcome evidence, reconcile compute and service charges, and transfer value under the agreed rule. Keep inference advisories distinct from attested evidence.

WHAT THE CODE CURRENTLY GUARANTEES LESS STRONGLY THAN THE PROSE

Context and usage are optional. v1 inference receipts are described as unsigned advisories, and the provider does not automatically attach a universal attested receipt. Edge has a domain capability gate, while policy enforcement depends on the host path and configuration. [S1, S2, S4, S5]

Reference composition, not a claim that the full approval, billing and signing workflow is automatically wired by either package.

COMMERCIAL INTERPRETATION

Six new ways to create useful services

These are product-design implications of the interfaces, not claims that Totem already sells the vertical products.

OPPORTUNITY	WHAT THE TWO PACKAGES ADD
Knowledge around an asset	Passport and service copilots; private policy, contract and evidence workspaces. Combine OCR, embeddings, retrieval and language with source permissions.
Accessible field operation	Transcription, translation and spoken guidance for technicians, drivers, farmers, customers and members. Link the conversation to qualified work and receipts.
Governed agent procurement	Assistants that prepare invoices, RFQs, service bookings and treasury proposals, with an independently enforced mandate for compute and downstream spending.
Embodied task services	Language/VLA task proposals and generative previews connected to robot identity, authority, bounded controllers and agreed completion evidence.
Creative economic services	Licensed generation, model-enabled features and commissioned media with acceptance, provenance records and settlement.
AI infrastructure operations	Metered inference, local knowledge nodes, model-fleet administration and cooperative model services built around provisioned runtime resources.

A practical sequence for exploration

Begin with source-linked assistants and field workflows whose outputs receive review. Add paid inference and agent procurement once identity, metering and authority are reliable. Explore embodied execution and simulation-led decisions with domain-specific engineering validation. This sequencing is an editorial assessment of integration complexity, not a market or delivery forecast.

All 36 domain assessments follow. Original crossovers retain IDs 01-40; new intelligence-led concepts are numbered 41-52.

Payments, wallets & everyday digital money

A common transaction layer for people, businesses, software agents and machines.

USECASE UNIVERSE

Everyday money

Person-to-person, person-to-business and business-to-business payments; merchant and web checkout; QR requests, payment links and invoices; tips and donations; payroll and contractor settlement.

Recurring and granular services

Micropayments, pay-per-click and pay-per-request services; subscriptions, metered billing and prepaid balances; membership dues, association fees and DAO contributions.

Conditional value

Deposits and refundable deposits; escrowed purchases; conditional and timelocked payments; revenue shares, royalties and distribution workflows.

Machine and agent commerce

Machine-generated invoices; machine-to-human, human-to-machine and machine-to-machine payments; agent-initiated purchases under delegated wallet policy.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR can extract an invoice, an LLM can explain the charge, and retrieval can compare it with an approved purchase order. Speech, translation and spoken responses add accessible checkout and multilingual support. The useful new product is an invoice-to-payment assistant that prepares a structured intent with source references, amount, asset and recipient for independent approval.

COMPOSITION PATTERN

OCR / RAG / LLM → validated payment intent → wallet policy and authority → signing → settlement receipt.

Boundary. The model must not infer a trusted payee or signing permission from invoice text. Confirm recipient, amount and asset against authoritative records.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

DEX, swaps & Web3 capital markets

The fabric supports capital-market applications and the custody, policy and settlement beneath them.

USECASE UNIVERSE

Exchange

Token swaps, routed and cross-token Omnia swaps, atomic HTLC swaps, peer-to-peer exchange, OTC and RFQ markets; broker and aggregator front ends, DEX interfaces and liquidity routing.

Liquidity and trading

Liquidity-provider marketplaces, liquidity bonds and provider reputation; trading terminals, portfolio and token-discovery apps, analytics; agent trading, treasury rebalancing and basket strategies under mandates.

Issuance and distribution

Token launches and distribution tools, launchpads, community sales, crowdfunding and crowdinvesting; membership, revenue-share and utility tokens; secondary asset-token markets.

Economic access

Token-gated commerce and services; policy-controlled treasury execution and swaps. An AMM, order book, auction or other market mechanism remains application-specific.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

LLMs and retrieval can explain positions, research approved disclosures and draft RFQs. OCR can structure issuer documents; classification can triage exceptions; agents can propose rebalancing trades within a mandate. This makes a locally operated treasury copilot possible without putting a private key in the inference provider.

COMPOSITION PATTERN

Approved market data + retrieval → proposed trade → deterministic limits and slippage checks → authorised swap.

Boundary. Generated forecasts are not price oracles or investment guarantees. Market design, liquidity, execution controls and applicable obligations remain with the application.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Prediction, information & event markets

Observed physical events can become settlement inputs alongside conventional event data.

USECASE UNIVERSE

Public and financial events

Traditional event, sports, election, governance and asset-price markets; energy-price and demand forecasts; local weather outcomes and environmental conditions.

Industrial outcomes

Machine uptime, failure and maintenance outcomes; production output and factory throughput; vehicle failure, battery degradation and cold-chain breach prediction.

Mobility and capacity

Delivery arrival, port and traffic congestion, charger availability; data-centre load, GPU availability and network-bandwidth demand.

Measured outcomes

Renewable generation, grid demand and crop yield. Modbus, CAN, BACnet, LoRaWAN, OPC-UA, BLE and location evidence can contribute signed observations to an application-defined resolution process.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Classification and LLM-assisted analysis can interpret incident reports and combine them with structured telemetry. Retrieval can expose the exact market rules and relevant evidence; agents can quote or trade subject to policy. A local predictor becomes another evidence-producing service, with a disclosed model version and uncertainty, rather than the final judge of its own market.

COMPOSITION PATTERN

Telemetry + referenced model forecast → market participation → independent outcome evidence → dispute process → settlement.

Boundary. The application must define measurement windows, oracle independence, missing-data handling and disputes. A signed prediction does not establish the eventual outcome.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Omnia-native financial products

Programmable channels can carry economic state for services, communities, assets and fleets.

USECASE UNIVERSE

Eight baseline programs

Normal payment, counter, meter, HTLC, vault, treasury, membership and asset programs underpin bilateral payment, conditional exchange, lockboxes, spending caps, dividends and conservation of assets.

Service relationships

Pay-per-unit, second, byte, kWh or machine cycle; high-frequency micropayments, merchant routing, API and compute billing, sensor feeds, content streaming and game-state/payment channels.

Shared economics

Escrow, delayed release and savings; governance-controlled treasuries, membership and revenue-sharing channels, cooperative profit pools and local community currencies.

Network and liquidity structures

Multi-hop routing, cross-token swaps, channel factories, shared liquidity, virtual channels, fleets of low-value channels and splicing. VT XO commerce and statechain ownership transfer expand the wider off-chain design space.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Inference adds a new metered service: a channel can fund repeated queries, transcriptions, embeddings or generated media. An agent can select an approved model and propose the next request while a host enforces the commercial agreement. Intelligence usage fields provide an integration point for accounting, but measurement, pricing and reconciliation must be built around the actual runtime.

COMPOSITION PATTERN

Service tariff + reserved budget → inference run → final measured usage → commercial receipt → channel update.

Boundary. Usage is optional and best-effort in the adapter. Importing qvac does not create a billing engine or guarantee a hard compute-spend cap.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 05 OF 36

Real-world assets & asset financialisation

An asset's legal and economic record can be continuously informed by its operating evidence.

USECASE UNIVERSE

Asset classes

Real estate, infrastructure and equipment; solar farms, batteries, chargers, telecoms, data centres and GPU clusters; industrial machines, vehicle fleets, aircraft and marine assets.

Claims and cash flows

Commodities, inventory and warehouse receipts; receivables and invoice finance; IP, patents, licences, royalties and revenue-backed tokens; rent, interest, dividends and profit sharing.

Lifecycle

Tokenisation, fractionalisation, audit, distribution, transfer, redemption and disposal; custody and valuation records; lockups, conditioned transfers, secondary markets and decommissioning.

Living asset evidence

Operating hours, output, energy, utilisation, maintenance, faults, revenue and location; ownership, firmware and compliance status linked to the asset record.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR and retrieval can turn asset files into a searchable diligence workspace; LLMs can explain cash-flow changes against maintenance records. Models can propose health or residual-value indicators from appropriately prepared inputs. Investors and operators gain a copilot for a living RWA, while source documents, sensor attestations and legal rights remain distinguishable from model-derived assessments.

COMPOSITION PATTERN

Asset passport + revenue and telemetry → attributed analysis → valuation or financing proposal → governed distribution / transfer.

Boundary. Token records do not themselves establish enforceable title. AI valuations and risk scores need validation; RWA policy templates remain experimental unless separately verified.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Digital Product Passports

A passport can become a product's persistent identity for operation, control, service and economics.

USECASE UNIVERSE

Product families

Industrial machinery, automotive and EV batteries; consumer electronics, medical devices and pharmaceutical batches; food, luxury goods, textiles and construction materials.

Infrastructure and components

Buildings and components, solar panels, wind turbines, aviation and maritime equipment, agricultural machinery, robots and critical-infrastructure devices.

Identity and assurance

Manufacturer, model and serial identity; ownership, custody, location, installation and commissioning; certification, calibration, authorised parts, firmware versions and software hashes.

Lifecycle and economics

Maintenance, repair, warranty and safety inspections; environmental characteristics and carbon footprint; usage and transfer history, resale, recycling and disposal. Passport state can gate operation, updates, servicing and market access.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A passport copilot combines OCR for legacy certificates, embeddings and retrieval for manuals, and language or voice interfaces for technicians and buyers. It can explain what changed, find the applicable repair procedure and propose a service order. Generated summaries should point back to immutable evidence versions rather than silently becoming the authoritative passport.

COMPOSITION PATTERN

Passport evidence → authorised retrieval → service recommendation → qualified operator approval → action and payment receipts.

Boundary. Extracted text is a claim until checked against its source and issuer. The assistant cannot confer certification, ownership or operational permission.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Certification, accreditation & compliance as code

Evidence, issuer trust, revocation and attributes can be evaluated before a right is exercised.

USECASE UNIVERSE

Products and sites

Industrial, safety, ISO-related, food, organic, fair-trade and renewable-energy evidence; inspection and calibration certificates; medical devices, buildings, electrical systems and roadworthiness.

People and suppliers

Professional licences, operator and technician qualifications, driver permissions; vendor, supplier and contractor approval; aviation and maritime maintenance authorisations.

Trade and institutional controls

Regulatory and environmental permits, permit-to-work, export/import and customs workflows; KYC, AML, investor attributes and sanctions policies; supply-chain due diligence and corporate controls.

Layered authority

Manufacturer, regulator, owner, site and operator rules can coexist. Schema validation, issuer verification, revocation and attribute proofs provide the intended template pattern for automated gates.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR and classification can route incoming certificates; retrieval can assemble the current policy and evidence pack; translation can support cross-border review. LLMs can draft an exception explanation or identify missing documents. Deterministic schema, issuer, date and revocation checks then decide whether the submitted proof satisfies the applicable rule.

COMPOSITION PATTERN

Document extraction → source-linked evidence pack → deterministic compliance checks → authorised access or escalation.

Boundary. A model's interpretation is not a legal determination or certification. Maintain authoritative policy versions, qualified review and jurisdiction-specific implementation.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Supply chain, trade & logistics

Provenance and physical condition can connect delivery obligations to title, finance and settlement.

USECASE UNIVERSE

Provenance

End-to-end chain of custody, batch tracking and serialised goods; food, pharma, electronics and conflict minerals; luxury authentication, component provenance and genuine spare parts.

Condition and movement

Cold chain, temperature excursions, humidity, shock and vibration; door-open and reefer monitoring; proof of pickup and delivery, warehouse custody and inventory.

Trade documents

Bills of lading, electronic title, customs and multi-authority border approval; import/export records, quality checkpoints, acceptance tests and supplier compliance.

Commercial consequences

Reorder triggers, freight escrow, milestone payments, conditional invoices, penalties and SLAs; cargo insurance, trade and supply-chain finance; receivables based on proven delivery.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR can structure trade documents; translation can assist border and carrier handoffs; retrieval can reconcile contractual tolerances with signed temperature history. A logistics copilot can propose a claim, reorder or invoice release and explain the evidence. Intelligence improves interpretation around the established sense, proof, policy, action and settlement loop.

COMPOSITION PATTERN

Signed shipment history + document retrieval → exception proposal → custody and policy checks → trade action / conditional payment.

Boundary. The source sensor, custody chain and document issuer still require verification. Extraction confidence must not automatically transfer title or release disputed funds.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 09 OF 36

Industrial automation & smart manufacturing

The fabric connects factory observations, operating permissions, bounded actions and commercial outcomes.

USECASE UNIVERSE

Integration and evidence

PLC monitoring and control, SCADA, production lines, robot cells, CNC and process evidence; Modbus, OPC-UA, MQTT, CAN, ROS2 and gRPC integration; industrial sensor networks.

Reliability and quality

Condition monitoring, predictive and preventive maintenance, uptime SLAs and performance contracts; quality assurance, proof of production and completed work; safety interlocks and controlled recipes.

Equipment commerce

Pay-per-part and pay-per-cycle, Machine-as-a-Service, rental and leasing, feature licensing; capacity and spare-parts markets, maintenance escrow and warranty-conditioned servicing.

Governed operations

Technician authority, regulator and auditor access, multi-site policy, industrial data markets, factory cooperatives and DAOs; revenue-generating autonomous equipment.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A local operations copilot can retrieve manuals and prior faults, transcribe a technician's observations, classify available inputs and propose a maintenance work order. Where a suitable model and supported runtime exist, VLA can contribute proposed tasks for robotic cells. Intelligence turns factory records into usable recommendations; the action layer still bounds what the machine can execute.

COMPOSITION PATTERN

Telemetry + maintenance knowledge → diagnosis hypothesis → authorised work order → bounded action → measured outcome and settlement.

Boundary. Model availability does not certify a control loop. Preserve engineered interlocks and validate domain-specific models against actual equipment and fault conditions.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Robotics & Physical AI

Intelligence broadens the task-proposal layer while Totem separates proposal, authority, actuation and settlement.

USECASE UNIVERSE

Embodiments

Industrial and warehouse robots, delivery and agricultural robots, inspection and service robots, AMRs and arms; drones, autonomous vehicles, construction and mining machines, remote manipulators and machine vision.

Autonomous services

Robot-as-a-Service, task and labour marketplaces, pay-per-task; charging, maintenance and route negotiation; tolls, procurement and energy purchases.

Fleet operation

AI-controlled buildings, factories and microgrids; fleet budgets, human approval and regulator constraints; DAO-governed fleets and cryptographically bounded agency.

Accountability

Device, app, agent, sensor, robot, gateway and service identities; explicit operating envelopes; verifiable action receipts connected to service and payment obligations.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

QVAC's VLA adapter exposes inference and embodiment/preprocessing helpers; world exposes scene creation and stepping. They create integration opportunities for language-guided task proposals, simulated previews and local operator assistance. ASR, translation and TTS add a hands-free interface. None of these wrappers is itself a robot controller or proof that a proposed action is safe.

COMPOSITION PATTERN

AI proposes → policy evaluates → authority approves → governed key signs → bounded controller acts → receipt proves → Omnia settles.

Boundary. Keep runtime inference outside the signing boundary. Validate task outputs and enforce real-time safety in the controller; generated world scenes are not validated physics.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Energy, utilities & climate

Metered resources can participate in markets, incentives and collective governance.

USECASE UNIVERSE

Electricity markets

P2P sales, prosumers, rooftop and community solar, virtual power plants and microgrids; islanded-grid governance, balancing, flexibility, demand response and load shedding.

Tariffs and storage

Dynamic tariffs, real-time pricing and net metering; battery arbitrage and charge/discharge mandates; home battery agents, industrial energy management and grid-service compensation.

Charging and instruments

EV charging, roaming, reservations, pay-per-kWh and charger channels; REC issuance, trading and retirement; meter-backed carbon instruments, carbon MRV and efficiency credits.

Other utilities and ownership

Water, gas and district heating, shared infrastructure and utility incentives; energy cooperatives and DAOs, community and machine-owned energy assets.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Local agents can retrieve tariffs, explain meter anomalies and propose battery or demand-response schedules. Forecasting can be composed from suitable models and data, with intelligence providing invocation contracts rather than a dedicated energy forecaster. Voice interfaces can explain household commitments; policy can cap expenditure, reserve backup capacity and require approval for changed operating envelopes.

COMPOSITION PATTERN

Meters + tariff knowledge + model proposal → grid and owner policy → bounded dispatch → verified delivery → settlement.

Boundary. Predictions do not replace metering. Credit eligibility, market admission and grid constraints require their own validated rules and recognised evidence processes.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Buildings, real estate & hospitality

Access, facilities, utilities and property economics can share a governed evidence layer.

USECASE UNIVERSE

Building operation

Smart buildings, HVAC control and optimisation, lighting, security, occupancy and environment sensing; BACnet, Matter and BLE provide relevant local integration patterns.

Access and services

Smart locks and elevators, workspace and equipment booking; tenant utilities and services, shared energy, parking and charging; smart apartments, hotels and vacation rentals with time-limited credentials.

Facilities and assurance

Maintenance, contractor permit-to-work, building compliance and fire/safety records; asset maintenance passports and energy-performance contracts.

Property economics

Tokenised and fractional property, rental-income distribution, condo/HOA and cooperative-housing governance; building treasuries and community energy. Offline access can use locally verifiable mandates where implemented.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A building assistant can answer tenant questions from approved records, translate requests and propose maintenance visits. Operators can ask about energy use or retrieve the procedure for an asset on site. Language-guided access requests become structured proposals checked against current credentials, time windows and site policy; conversation alone never grants entry.

COMPOSITION PATTERN

Tenant request or building evidence → authorised retrieval → service/access proposal → site policy → bounded action and receipt.

Boundary. Do not treat voice recognition as identity. Local inference, offline access verification and later settlement are separate availability requirements.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Automotive, mobility & transport

Vehicle evidence can govern access, usage economics, maintenance and autonomous procurement.

USECASE UNIVERSE

Vehicle lifecycle

Identity and Digital Product Passports, telemetry, mileage and usage proofs, maintenance and fleet management; automotive, heavy machinery, construction and agricultural vehicles.

Usage economics

Usage-based leasing, finance and insurance, pay-per-mile cover, rental and car sharing; micromobility and bike/scooter payments; monetisation of vehicle data.

Movement and energy

EV charging and roaming, parking, tolls and congestion charges; fleet fuel and energy procurement; autonomous vehicle payments and robotaxi settlement.

Transport infrastructure

Logistics fleets, rail assets, maritime and port equipment, aviation component passports; drone services and delivery, airspace and landing-slot markets. CAN and location proofs supply relevant evidence interfaces.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A local fleet copilot can summarise fault reports, retrieve servicing instructions and translate driver observations. Agents can propose charging, parking and repair purchases within fleet mandates. VLA provides an additional inference interface for compatible embodied systems, while OCR can ingest repair and component records into the vehicle passport.

COMPOSITION PATTERN

Vehicle evidence + driver request → service or route proposal → fleet budget and geofence policy → approved transaction / controller action.

Boundary. The adapter does not supply autonomous-driving certification or a navigation stack. Vehicle safety, privacy and location integrity need independent controls.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 14 OF 36

Agriculture, food & environmental systems

Low-connectivity sensing can support farm decisions, resource markets and environmental evidence.

USECASE UNIVERSE

Farm operation

Soil moisture, weather and crop sensing, precision irrigation and greenhouse automation; livestock tracking and health, farm machinery, energy, water allocation and payments.

Food provenance

Inputs and seed provenance, fertiliser and pesticide compliance, harvest and food-chain traceability; cold storage and transport, agricultural equipment leasing and farm cooperatives.

Risk and forecasting

Crop and parametric weather insurance, crop-yield prediction markets; evidence-linked production and service agreements.

Environmental systems

Carbon farming and soil-carbon MRV, biodiversity, forestry, water and air-quality monitoring; environmental credits, conservation payments and compliance. LoRaWAN provides a relevant rural sensing route.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A gateway-hosted assistant can retrieve agronomic procedures, transcribe field notes, translate instructions and propose irrigation or service tasks. Suitable models can analyse prepared sensor inputs, with provenance carried into the proposal. This extends the sensor-data market into paid interpretation, while respecting the distinction between a small sensor and a gateway capable of running a model.

COMPOSITION PATTERN

Signed rural observations → gateway inference + approved knowledge → farmer policy → irrigation/service action → evidence-based payment.

Boundary. Do not assume native models fit on LoRaWAN endpoints. Agronomic models and environmental claims require site-specific validation and recognised measurement methods.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Healthcare, medtech & life sciences

Identity, consent and regulated evidence can coordinate devices, research and access to sensitive information.

USECASE UNIVERSE

Devices and monitoring

Medical-device identity, certification and operating policy; maintenance, calibration, adverse events and post-market surveillance; wearables, remote monitoring and hospital equipment passports.

Consent and research

Patient, research and insurance-data consent and revocation; clinical-trial protocols, compliance and payments; patient-controlled health-data markets.

Care and access

Prescription validation, pharmacy workflows, provider and staff credentials; health-record access and sharing agreements, cross-border access and GDPR/HIPAA-style control patterns.

Supply and finance

Pharma and drug-batch provenance, vaccine cold chain, medtech asset finance and regulated data marketplaces; cross-domain trust for devices used outside their manufacturer's jurisdiction.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Local document understanding, transcription, translation and consent-filtered retrieval can assist administrative and research workflows. A staff assistant can find an approved device procedure or organise trial evidence without automatically sending it to a cloud model. BCI transcription methods exist in the ASR adapter, but their presence establishes an API surface, not clinical readiness or a new diagnostic product.

COMPOSITION PATTERN

Authorised records + consent scope → bounded retrieval / summarisation → qualified review → recorded access or workflow action.

Boundary. No clinical efficacy, diagnosis or compliance guarantee is implied. Validate intended use, model performance, deployment privacy and human oversight for the actual setting.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Telecommunications & connectivity

Connectivity can be discovered, purchased, metered and governed as a machine service.

USECASE UNIVERSE

Access markets

Paid Wi-Fi hotspots, pay-per-byte internet, community broadband, bandwidth markets and mobile backhaul sharing; IoT connectivity and LoRaWAN gateway services.

Commercial relationships

Roaming and network-SLA settlement, service-provider bonding, machine subscriptions and autonomous network procurement.

Infrastructure

Tower sharing, network capacity, satellite connectivity and ground-station access; mesh, peer-to-peer and decentralised connectivity networks.

Service interfaces

API access, gRPC and edge-service billing, secure discovery and machine-to-machine connectivity purchasing. MachinePay's hotspot pattern anchors the pay-per-byte service model in the supplied scope.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

An edge service can combine connectivity with an approved local model, such as transcription or translation at a community node. Operators can retrieve procedures and analyse logs; software agents can compare service offers and propose capacity purchases. QVAC model and system operations create a management seam for inference-capable gateways alongside existing network economics.

COMPOSITION PATTERN

Service discovery + local resource information → approved connectivity / inference purchase → metered service → SLA and payment receipts.

Boundary. Network routing, radio functions and roaming agreements are not delivered by the intelligence adapter. An advertised domain is not proof of available capacity.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Compute, AI infrastructure & data centres

The new packages turn AI execution itself into an explicit service that can join the economic fabric.

USECASE UNIVERSE

Compute services

GPU, CPU and edge compute markets; inference, pay-per-query, GPU-second and container services; serverless and batch workloads, rendering, storage and combined bandwidth/compute.

Capacity and operations

Reservations, spot markets and capacity futures; data-centre energy procurement and cooling optimisation; autonomous compute brokers and distributed Physical AI infrastructure.

AI commerce

Agent-to-agent APIs, agents procuring compute, dataset markets, model and feature licensing, AI subscriptions and specialised inference services.

Runtime operations

QVAC surfaces model load/unload, registry lookup, fit assessment, asset download, resource inspection, logs and cancellation. These can support a managed local-model service when runtime and hardware are provisioned.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Intelligence supplies a provider contract; qvac supplies a concrete adapter with streaming and model lifecycle interfaces. A host can offer an authorised inference endpoint, expose supported models and quote a tariff. This opens paid local inference, private enterprise inference appliances and managed model fleets, with Omnia or other settlement composed around measured usage.

COMPOSITION PATTERN

Model/service offer → identity and budget checks → provisioned runtime → final usage collection → invoice or channel settlement.

Boundary. No automatic distributed scheduler, GPU marketplace, guaranteed usage attestation or token-to-price engine is supplied. Benchmark fit, latency, concurrency and cost on target hardware.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Data markets, telemetry & digital twins

Buyers can pay for governed access to observations and, now, for interpretations built from them.

USECASE UNIVERSE

Data products

Sensor, industrial, vehicle, energy, environmental, farm, building, health and location feeds; geospatial and raster imagery; machine-performance and digital-twin data.

Access and pricing

Pay-per-reading and query, subscriptions and time-, category- or purpose-limited licences; resale policy, data escrow and controlled release.

Evidence and privacy

Provenance, quality proofs and access receipts; privacy-preserving sharing and ZK claim integration where implemented; commercial telemetry licensing.

Derived assets

AI-training data licensing and machine-generated IP; machine-state summaries, semantic search and licensed derived datasets, with lineage back to the underlying observations.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Embeddings and RAG make authorised telemetry records and maintenance documents searchable; classification and generation can create derived reports or semantic feeds. A buyer can purchase an interpretation as well as raw readings. The licence must distinguish source access, embedding storage, model use and resale of derived outputs; a model summary should retain the relevant evidence references.

COMPOSITION PATTERN

Licensed source data → consent-filtered retrieval / inference → attributed derived product → access receipt → payment.

Boundary. RAG is not a complete digital-twin engine. Generated world scenes and summaries do not certify physical state, data quality or permitted reuse.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Geospatial, mapping & location-aware systems

Location, spatial and raster evidence can condition physical access, markets and claims.

USECASE UNIVERSE

Movement and access

Proof of location, route and delivery; geofenced access and payment, vehicle and drone route compliance, maritime routes, cargo tracking and equipment location.

Rights and charging

Workforce/site attendance, insurance location evidence, tolls, parking and congestion zones; time- and place-bounded service rights.

Land and environment

Land-use, forestry and agriculture mapping; environmental monitoring, remote-sensing and satellite/aerial-image provenance.

Inspection evidence

Surveys, construction progress, property inspection, disaster assessment and insurance claims. Separate location, spatial and raster proof families form the scope's evidence substrate.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR can ingest survey records; retrieval can connect a site to approved permits and maintenance history. Suitable classification or multimodal models can propose interpretations of location-linked material; language interfaces can explain geofence constraints. The model-derived description should remain attached to the original image, timestamp, location claim and processing version.

COMPOSITION PATTERN

Spatial evidence + relevant documents → model-assisted interpretation → geofence / permit checks → authorised service or claim workflow.

Boundary. Image interpretation does not prove location or survey accuracy. Specialist GIS, geodesy, sensing validation and authoritative boundary data remain application responsibilities.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Privacy, consent & personal data

Consent can govern both access to data and the purposes for which intelligence may process it.

USECASE UNIVERSE

Consent controls

User-controlled grants and receipts, revocation, purpose and data-category limits, processor authorisation and retention policies; auditable data-access logs.

Subject workflows

Access, rectification, erasure and portability; cross-provider portability, data escrow and conditional or time-locked disclosure.

Domain applications

Health, financial and automotive consent; industrial telemetry licensing, privacy-controlled markets and AI dataset permissions.

New processing boundaries

Source documents, chunks, embeddings, prompts, outputs, model adaptations and logs each need a defined retention and access policy. Local execution can reduce data movement when configured appropriately.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Private retrieval and local inference make consent-aware assistants possible for people, enterprises and cooperatives. The host should apply permissions before retrieval, minimise the material sent to a model and record the processing purpose. QVAC's RAG workspace and embedding deletion methods provide useful operations; the application must propagate revocation through every derived store and downstream recipient.

COMPOSITION PATTERN

Consent + principal → filtered retrieval → approved model invocation → access / usage record → retention and revocation workflow.

Boundary. Local execution is not automatic end-to-end privacy. Downloads, plugins, telemetry, caches, logs and backups require explicit deployment choices and controls.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Identity, credentials & access control

Portable claims and delegated authority can govern people, devices, agents and services.

USECASE UNIVERSE

Identity subjects

Humans, devices, machines, agents, organisations and services; delegated identities, recovery identities and rotatable controllers.

Credentials

Employee, technician, professional, supplier, member and visitor credentials; badges, hotel credentials and equipment-operator permissions.

Digital and physical access

API/service authentication, dApp sign-in and proof of control; role- and capability-based access, time-bounded authority and zero-trust device access.

Operational scope

Access-control patterns connect operator identity and policy roots to Modbus, OPC-UA, BACnet, ROS2 and Matter operations. Intelligence context can carry agent, principal, proposal and run identifiers.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

An assistant can explain entitlements, locate credentials and turn a natural-language request into a structured access proposal. OCR can assist credential intake, and translation can help cross-border service desks. Context fields make correlation possible, but they are optional metadata at the provider interface; the host must authenticate the principal and enforce the actual permissions.

COMPOSITION PATTERN

Authenticated principal → entitlement-filtered assistant → structured request → independent credential / authority verification → access.

Boundary. Prompt text, voice and self-declared context are not identity proofs. Validate the domain, operation, parameters and authorisation together before dispatch.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Device security & software lifecycle

Policy can govern a device from commissioning through software changes to secure retirement.

USECASE UNIVERSE

Identity lifecycle

Commissioning, ownership enrolment and key provisioning; rotation, recovery and credential/operator revocation; emergency policy revocation and incident-response authority.

Software governance

Firmware signing and deployment, increasing-version anti-rollback policy, software provenance and firmware hashes; manufacturer and owner approvals.

Operational protection

Secure commands, zero-trust access, machine configuration governance and audit receipts; transfer, wipe, decommissioning and disposal.

AI runtime extension

Model, runtime, prompt, retrieval corpus and plugin versions can become governed deployment inputs. Model inventory and resource operations provide a seam for fleet oversight.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Retrieval can support incident response and firmware review; an assistant can propose a rollback-safe update plan. QVAC model loading, unloading, registry queries and system logs can be wrapped in the same approval discipline used for equipment configuration. A model or plugin change should be treated as a change to operational behaviour, with an identified owner and validated release.

COMPOSITION PATTERN

Change evidence + model/runtime inventory → proposed update → manufacturer / owner / site policy → deployment receipt.

Boundary. The adapter is not an OS sandbox, model-signing system or malware scanner. Direct provider and raw SDK access must remain outside untrusted callers' reach.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Warranty, servicing & aftermarket

The product record, service authority and measured repair outcome can support an aftermarket economy.

USECASE UNIVERSE

Cover and conditions

Manufacturer and extended warranties; usage- and maintenance-conditioned cover, uptime guarantees, machine-health subscriptions and warranty insurance.

Service execution

Service and repair history, approved parts and technicians; booking, maintenance escrow and payment after verified repair; predictive and outcome-based contracts.

Parts and reuse

Spare-parts markets and provenance, refurbishment certificates, residual-value estimation, used-equipment and vehicle markets.

Lifecycle continuity

Warranty transfer with ownership; maintenance evidence carried by the product passport and used in subsequent servicing, financing or resale.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A service copilot can identify the applicable manual, retrieve warranty conditions, transcribe an inspection and propose the parts and technician required. The customer can receive a source-linked explanation in their language. Post-repair evidence can be organised for review and escrow release, reducing administrative work while preserving independent acceptance criteria.

COMPOSITION PATTERN

Fault evidence + passport retrieval → service proposal → approved parts / technician → repair evidence → warranty update and payment.

Boundary. A generated diagnosis is a hypothesis. Warranty entitlement, technical suitability and successful repair need the agreed evidence and authorised acceptance.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Insurance & risk

Evidence, policy and programmable settlement can compose insurance products without implying an insurance package.

USECASE UNIVERSE

Physical risks

Parametric crop, weather and cargo cover; cold chain, vehicle telematics and pay-per-mile; equipment breakdown, downtime, battery and renewable-generation performance.

Service and operational risks

Building performance, SLAs, delivery and product warranties; cyber-physical device and escrow risk; proof-triggered claims and parametric payouts.

Distribution and pooling

Peer-to-peer mutuals, DAO pools, machine mutuals and microinsurance; event-, trip- and usage-specific cover.

Duration innovation

Minute-, hour- and day-duration machine cover, automatically expiring policies and session-bundled premiums; evidence from the insured asset linked to its policy and settlement.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR and retrieval can prepare policy and claims files; transcription and translation can improve field reporting. Models can propose risk indicators from suitable inputs, explain exclusions and triage claims. Agents can request short-duration cover before a task, with underwriting, terms and premium approval enforced independently of the assistant.

COMPOSITION PATTERN

Task and asset evidence → risk / cover proposal → underwriting and authority → premium → objective claim evidence → settlement.

Boundary. Totem does not become an insurer or actuarial model by composition. Coverage, reserves, licensing, disputes and fair claims handling require application-specific design.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Legal, notarisation & digital instruments

Evidence and delegated signing can coordinate instruments, approvals and conditional agreements.

USECASE UNIVERSE

Records and instruments

Document notarisation, proof of existence and timestamps; contracts, deeds, affidavits, certificates, wills and powers of attorney; witness workflows and evidence chains.

Conditional execution

Legal escrow, conditional contracts and multi-party agreements; delegated signing authority and procurement approvals.

Institutional decisions

Corporate and shareholder resolutions, board approvals and intellectual-property records.

Recognition

Cross-border instruments, mutual-recognition frameworks and regulatory recognition. Legal/notarial templates form part of the original Recursive MAST scope.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

OCR, retrieval and translation can assist document intake, clause comparison and evidence assembly. An assistant can prepare a resolution or explain a proposed mandate using approved sources. The resulting text becomes a draft for authorised review; only the established signing and recognition process can turn it into the relevant instrument.

COMPOSITION PATTERN

Source instruments → controlled retrieval / drafting → qualified review → identity and authority checks → signed record.

Boundary. A hash or model summary does not confer legal validity. Jurisdiction, capacity, witnessing, formalities and recognition remain separate requirements.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Governance, DAOs & cooperatives

Collective decisions can govern treasuries, digital services, physical assets and operating mandates.

USECASE UNIVERSE

Organisations

Token, membership, investment, RWA, infrastructure and energy DAOs; data, worker, consumer, housing, farm, local-energy and machine-owner cooperatives.

Decision models

Linear, quadratic and liquid voting; expert delegation, minority protection and member admission/removal; protocol and fleet governance.

Collective resources

Treasuries, grants committees, public-goods funding, participatory budgets and community funds; governance as an alternative to simple multisignature administration.

Execution

Policy updates, treasury spending and device/fleet changes; passed proposals expressed through authority mandates, including governance-triggered physical actions.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A governance copilot can retrieve the constitution and prior decisions, summarise proposals, translate debate and draft execution intents. Members can query the evidence behind an equipment or treasury decision. Intelligence makes participation and administration easier, while voting rules, quorum, minority protections and authority remain deterministic and inspectable.

COMPOSITION PATTERN

Constitution + proposal evidence → member-facing analysis → formal vote → authority mandate → bounded treasury or physical action.

Boundary. Generated summaries must not replace the proposal of record. Models do not establish quorum, count votes or gain executive authority through a conversation.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Network States & digital polities

A digital polity can govern membership, shared resources and physical services through a common fabric.

USECASE UNIVERSE

Membership and constitution

Citizenship or membership identity, admission and revocation, constitutional policy roots, local chapters, delegated authority and cross-community credentials.

Institutions

Liquid and quadratic governance, referenda, councils and working groups; service-provider registries, certification, appeals, rulings and policy checkpoints.

Public economics

Treasuries, dues, local currencies, grants and procurement; land/property records, RWA and community infrastructure ownership.

Physical polity

Cooperative energy, broadband and compute; property access, robots and machine-operated public services; cross-jurisdiction trust and mutual recognition; bounded AI-supported administration.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A private institutional knowledge service can answer questions about rules, records and services in multiple languages. Agents can prepare procurement or public-goods proposals, and local inference nodes can be community infrastructure. The new opportunity is a polity with an accountable administrative assistant and governed compute services, alongside its existing membership and economic institutions.

COMPOSITION PATTERN

Member identity + constitutional retrieval → administrative proposal → institutional process → delegated mandate → public-service receipt.

Boundary. Software membership is not sovereign recognition. Rights, due process, real-world institutions and the limits of delegated authority must be explicitly established.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Public sector & smart cities

Public services can connect permits, infrastructure evidence, procurement and accountable execution.

USECASE UNIVERSE

Administration

Municipal identity, business and operator licences, permits and building approvals; land/property records, public asset inventories and public audits.

Infrastructure

Traffic, tolls, parking and public charging; sensor networks, air/water quality, waste, recycling, utilities and maintenance.

Public spending

Procurement and contract milestones, grants, benefit-condition proofs and participatory budgets.

Community and emergencies

Public housing governance, community energy and emergency-authority delegation; services can use the fabric without decentralised governance.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Document extraction and multilingual retrieval can improve permit intake and public-service navigation. Staff copilots can assemble inspection evidence and draft procurement milestones; agents can propose maintenance based on reported conditions. Private deployment can help keep sensitive administrative material within a controlled environment when data flows are configured accordingly.

COMPOSITION PATTERN

Applicant or infrastructure evidence → assisted intake → published rule checks and review → authorised service / procurement action.

Boundary. Models should not silently determine entitlements or enforcement. Preserve reasons, appeals, accessibility, lawful authority and auditable human responsibility.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Retail, vending & unattended commerce

A machine can verify a right to service, deliver it and record the economic outcome.

USECASE UNIVERSE

Retail

Vending, kiosks, smart shelves, automated retail and unstaffed stores; loyalty and memberships.

Self-service equipment

Pay-per-use appliances, laundry, printers/copiers, coffee and entertainment machines, car washes and public showers.

Access and rental

Charging and rental lockers, luggage storage, tool and bike/scooter rental, access gates and parking.

Disconnected operation

Offline-capable vending and payment queues where supported; locally verified entitlements with explicit reconciliation and failure handling.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Local speech, translation and TTS can make an unattended machine easier to use. A product or service assistant can answer questions from an approved catalogue and prepare a purchase request. Classification and OCR can assist support or inventory intake where suitable models exist. The transaction and physical release remain bound to a validated entitlement.

COMPOSITION PATTERN

User request → approved catalogue / price → payment or entitlement check → bounded dispense / unlock → receipt.

Boundary. Local AI availability does not guarantee offline payment finality. Define refund, failed-dispense, stock and reconciliation behaviour independently of the assistant.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Machine-as-a-Service & servitisation

Equipment can be sold as an ongoing measured service with bundled operational obligations.

USECASE UNIVERSE

Usage units

Pay-per-cycle, part, hour, kilometre, hectare or use; industrial, agricultural, medical and vehicle services.

Entitlements and pricing

Feature services and firmware unlocks, performance- and outcome-based pricing, lease enforcement and manufacturer revenue shares.

Service assets

Robot-, drone-, generator-, battery-, HVAC-, compressor- and tool-as-a-Service; equipment leasing and capacity subscriptions.

Bundled obligations

Maintenance in the usage price and insurance for individual sessions; policy-conditioned service, escrow and feature licensing in commercial template patterns.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A service assistant can quote the appropriate plan, retrieve operating instructions and propose a session with the necessary insurance and maintenance status. Local models can support service triage and operator guidance. A manufacturer can offer paid intelligence as part of the machine service, separately metered and licensed where the business model requires it.

COMPOSITION PATTERN

Task request + asset passport → service proposal → entitlement / insurance / operating checks → session evidence → usage settlement.

Boundary. Usage price, machine output and inference cost are different meters. Enforce lease and safety rules through authorised control logic, not generated instructions.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Creator, IP & licensing markets

Rights, entitlements and settlement can support both existing creative assets and generated work.

USECASE UNIVERSE

Intellectual property

Patent tokens and royalties, software and machine-firmware licences, datasets and AI models; IP-backed finance and royalty-backed RWAs.

Creative rights

Content, music and media rights, image/video licences and digital-rights receipts; pay-per-view and revenue-share IP.

Entitlements

Feature licences, pay-per-use software, time- and geography-limited access, device-bound licences and enterprise entitlement systems.

New generated services

Commissioned text, images, speech, audio and video; licensed model adaptations and model-enabled feature subscriptions, with separately defined input and output rights.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

QVAC exposes diffusion/upscaling, audio and video generation, TTS, completion and fine-tuning interfaces. These can power a commission marketplace in which an agent requests work, an authorised provider produces it and escrow releases after agreed acceptance. The fabric can record the model and licence terms; it cannot decide ownership merely from who invoked the model.

COMPOSITION PATTERN

Rights and commission terms → authorised generation → provenance / licence record → acceptance → creator and provider settlement.

Boundary. Model, dataset, likeness and output rights require their own agreements. A generation receipt does not establish copyright ownership or originality.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 32 OF 36

Gaming, interactive apps & communities

Programmable state and economic relationships can support games, creators and digital communities.

USECASE UNIVERSE

Game state and assets

Multiplayer state channels, turn-based games, in-game payments, item ownership and player marketplaces; KISSVM state machines and Omnia primitives provide composable patterns.

Competition

Tournament escrow, skill competitions, prize pools, wagering and prediction games, subject to the application's rules and obligations.

Community economics

Membership, guild treasuries and DAOs, fan and creator tokens, fan memberships and event tickets.

Interactive media

Streaming micropayments, generated dialogue, local voice interfaces, creative tools and paid world or content services.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

LLM, speech and generative-media adapters can support dialogue, translation, narration and creative assets. World scene/step interfaces offer an experimental integration point for generative environments. Agents can purchase approved content or services. Authoritative game state, anti-cheat logic and financial rules should remain separate from stochastic model output.

COMPOSITION PATTERN

Player permission + content budget → inference / generation → application validation → authoritative state update → settlement.

Boundary. These are application patterns, not a shipped gaming SDK. Generated scenes are not an authoritative multiplayer state engine or certified simulation.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Education, skills & workforce

Credentials can connect learning, professional authority, job access and completion-based payment.

USECASE UNIVERSE

Learning records

Academic and professional credentials, skills, training completion, continuing education and apprenticeship records.

Access and lifecycle

Job-site and machine-use permissions, credential expiry and revocation, cross-border recognition.

Work qualifications

Safety training, industrial operator certification, medical qualifications and technician licences; employer authorisation and gig-worker credentials.

Learning economics

Pay-on-completion programmes and training escrow; enterprise training services, local tutors and licensed instructional content.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Retrieval can ground a tutor in approved course material; speech and translation improve accessibility; generated scenarios can support practice. A workplace assistant can find the training required for a task and prepare a learning plan. Assessment results and instructor decisions remain attributable to the recognised issuer before any new machine-use permission is granted.

COMPOSITION PATTERN

Learner identity + approved curriculum → tutoring / practice → validated assessment → issuer credential → policy-controlled work access.

Boundary. An assistant's judgement does not issue a recognised qualification. Keep training assistance, formal assessment and the authority to operate equipment distinct.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

Circular economy & end-of-life systems

Product evidence can continue through repair, resale, second life, material recovery and retirement.

USECASE UNIVERSE

Repair and reuse

Repairability and repair histories, reuse markets, refurbishment certificates and resale provenance; battery second life and parts harvesting.

Recovery

Recycling evidence, waste tracking and e-waste compliance; deposit returns, recycling incentives, product take-back and Extended Producer Responsibility workflows.

Materials

Material passports and construction-material reuse; carbon accounting for reuse and sustainability evidence.

Economic closure

End-of-life asset redemption, residual-value markets and recorded decommissioning, connecting DPP, lifecycle and provenance patterns.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

A local assistant can retrieve dismantling instructions, identify applicable recovery documentation and organise a refurbishment record. Suitable classification models can propose material or part categories; OCR can ingest legacy labels and certificates. Buyers can query the history behind a used asset, and agents can propose take-back or recycling transactions under the product's policy.

COMPOSITION PATTERN

Passport + physical inspection → recovery recommendation → authorised facility / method → verified recovery event → incentive or redemption.

Boundary. Generated material identification and environmental estimates need validation. A passport entry alone does not prove safe handling, recycling or a valid credit.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 35 OF 36

Physical infrastructure / DePIN

Infrastructure networks can coordinate identity, evidence, service discovery, governance and real service payments.

USECASE UNIVERSE

Observation networks

Sensors, weather, mapping and environmental networks, with evidence-linked service quality.

Digital infrastructure

Wireless, LoRa and Wi-Fi, compute, GPUs and storage; community and shared industrial infrastructure.

Physical services

Energy, charging, mobility, robotics, drones, machine services and smart buildings.

Network economics

Provider discovery and reputation, resource access, collective governance, payments and bounded actuation; incentives can relate to useful service rather than reward emission alone.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

QVAC gives an infrastructure node an optional local intelligence service, while intelligence keeps the host-facing contract provider-neutral. Networks can offer transcription, retrieval or other provisioned inference alongside connectivity and compute. Model/system interfaces can support operations; licensed datasets and local expertise can form additional services on the same economic fabric.

COMPOSITION PATTERN

Node identity + verified service offer → authorised job → provisioned inference / physical service → quality evidence → settlement.

Boundary. The packages do not automatically create a distributed AI network, proof of inference or honest-compute protocol. Service verification and marketplace coordination remain to be built.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

APPLICATION DOMAIN / 36 OF 36

Autonomous-agent economy

Software agents can buy, interpret and propose action across the digital and physical economy under bounded mandates.

USECASE UNIVERSE

Authority and budgets

Agent wallets, delegated spending, budgets and rate limits; explainable approvals, human escalation, receipts and DAO- or machine-controlled agents.

Digital procurement

Subscriptions, APIs, compute, data and swaps; agent-to-agent commerce, treasury management, bidding and marketplace participation.

Physical procurement

Travel, logistics, energy and maintenance purchases; agents operating equipment through separate authorised action surfaces.

Intelligence integration

Provider-neutral invocation and streaming, context correlation, usage fields and a QVAC adapter. Agent-policy TypeScript types include an inference intent and an optional inference receipt field.

INTELLIGENCE + QVAC / THE NEW CONTRIBUTION

Agents can now use an explicit intelligence seam to interpret documents, retrieve knowledge, converse, generate assets or propose embodied tasks. This supports a governed agent that can request compute as well as spend on downstream services. The host must bind identity and proposal context, authorise each meaningful action and keep signing keys out of the inference process's accessible surface.

COMPOSITION PATTERN

Mandate → authorised inference request → validated proposal → separate action approval → governed signature → action and economic receipts.

Boundary. Optional context and receipt fields are integration hooks, not automatic enforcement. The adapter supplies neither a complete agent planner nor an end-to-end autonomous execution policy.

Original application scope [S7]. Intelligence assessment [S1-S5]. Composable application patterns; runtime and domain validation required.

The original 40, strengthened

Original concepts and numbering are retained. Each entry incorporates the intelligence contribution.

01 **Micro-duration machine insurance**

A robot requests 30 minutes of cover before hazardous work. Identity, live telemetry and policy define the session; Omnia can settle the premium. Intelligence can prepare a cover request and explain exclusions, while underwriting and objective claims rules remain independent.

02 **Machine uptime prediction markets**

Participants trade a turbine's future availability. Intelligence can analyse approved histories and propose forecasts; the machine supplies outcome evidence. Market rules must separate prediction from resolution and define downtime, measurement windows and disputed readings.

03 **Predictive-maintenance insurance**

Verified health evidence informs a cover or maintenance proposal. Local retrieval and suitable models can explain anomalies against service history. Changes to price, cover or mandatory work follow agreed policy and underwriting rather than unreviewed generated diagnoses.

04 **Autonomous maintenance escrow**

A machine reports a fault; an assistant retrieves the manual and proposes a qualified technician and parts. Authority approves procurement, funds enter escrow, and post-repair evidence plus agreed acceptance releases payment.

05 **Machine-backed RWA with live revenue**

A charger, robot, solar inverter, vending machine or GPU cluster links operating and revenue evidence to asset-holder distributions. An asset copilot explains performance and maintenance impacts, keeping forecasts separate from realised cash flow and ownership records.

06 **Proof-backed receivables**

A verified delivery, machine job or service can support a receivable. OCR and retrieval reconcile purchase orders, completion evidence and invoice terms before an authorised financing request. Financing quality still depends on enforceable claims and counterparty acceptance.

07 **Dynamic equipment leasing**

Lease pricing and limits can respond to hours, location and health. Intelligence interprets service evidence and proposes a revised plan; contractual policy controls changes. The original meter, operating envelope and accepted lease terms remain authoritative.

08 **Usage-based warranties**

Warranty conditions refer to operating limits and certified maintenance. A passport assistant retrieves the relevant terms and highlights missing evidence. Deterministic checks and authorised review establish entitlement rather than a model's opinion of the machine's condition.

Original concepts [S7]; intelligence integration is an application-design inference from [S1-S5].

The original 40, strengthened

Original concepts and numbering are retained. Each entry incorporates the intelligence contribution.

-
- 09 Autonomous robot charging markets**
- Robots discover chargers, compare approved offers, open service relationships and settle charging. Intelligence can interpret task urgency and propose a purchase; verified compatibility, availability, fleet budgets and safe controllers determine execution.
-
- 10 EV-charger prediction and reservation markets**
- Short future charging windows can be forecast, reserved or traded. Models contribute demand estimates and agents prepare bookings; measured availability, reservation rights and explicit cancellation rules govern delivery and settlement.
-
- 11 Edge-based energy prediction markets**
- Buildings, batteries and substations offer forecasts of generation or demand. Local inference can combine relevant records with meter inputs. Resolution uses agreed meter evidence and an independent market process rather than the forecasting model itself.
-
- 12 Autonomous microgrid economies**
- Assets buy and sell electricity under cooperative rules. Intelligence proposes schedules and explains trade-offs to members. Grid constraints, reserve requirements, owner authority and verified energy delivery govern dispatch and settlement.
-
- 13 Building flexibility markets**
- HVAC and other loads offer curtailment into demand-response markets. An assistant can retrieve comfort and equipment constraints and propose a bid. BACnet-linked measurements and an agreed baseline establish delivery; policy bounds the physical response.
-
- 14 Cold-chain finance**
- One verifiable temperature history informs shipment payment, invoice finance and insurance. OCR and retrieval align documents with excursion evidence, producing a reviewable exception proposal. Each commercial contract retains its own thresholds, rights and dispute process.
-
- 15 Autonomous cargo insurance**
- Cover attaches for a particular shipment leg and ends at an agreed custody transfer. Intelligence assembles the leg's documents and risk evidence; accepted terms, custody proofs and authorised premium payment define the actual coverage.
-
- 16 Data as a cash-flow asset**
- A sensor network earns per-reading or subscription revenue that can support asset-holder distributions. Intelligence adds licensed summaries or semantic queries as derived services. Revenue attribution, source licences and rights to derived outputs must be explicit.
-

Original concepts [S7]; intelligence integration is an application-design inference from [S1-S5].

The original 40, strengthened

Original concepts and numbering are retained. Each entry incorporates the intelligence contribution.

17 **Consent-paid personal data**

A person grants a defined purpose, dataset and period; approved use produces a receipt and payment. Consent-filtered retrieval and local inference can reduce unnecessary disclosure. Revocation must propagate to embeddings, caches and downstream processing.

18 **Carbon and REC evidence from equipment**

Meter evidence supports environmental-instrument workflows. Intelligence can assemble reporting packs and identify missing source material. Recognised methodologies, eligibility, independent checks and registry rules determine issuance and prevent double counting.

19 **Machine reputation bonds**

Machines and providers pair a financial bond with verifiable service history. Retrieval and models can summarise performance or triage complaints. Bond release or slashing follows published evidence and dispute rules, not a generated reputation score alone.

20 **Autonomous procurement**

A factory assistant identifies an approved spare part, checks provenance and proposes a purchase. Supplier credentials, amount limits and owner authority gate payment; receipt, installation and inventory records close the procurement loop.

21 **Smart spare parts**

A part carries its identity and passport. An assistant retrieves compatibility and servicing instructions; the machine's policy checks authorised part identity and installer qualifications. Generated identification never substitutes for provenance or safe installation checks.

22 **Governed firmware**

Manufacturer, regulator or fleet governance defines allowed firmware. Intelligence helps review release notes and prepare rollout proposals. Device-side signature, version and owner-policy checks enforce the decision, with model and plugin updates governed similarly where implemented.

23 **Pay-to-unlock machine capability**

A valid entitlement enables an equipment feature for a defined session or term. A local assistant can explain plans and prepare a purchase. Payment, licence verification and bounded control enforce the feature state.

24 **Dynamic residual-value markets**

Usage, servicing, damage, firmware and ownership evidence inform resale. A passport copilot organises the history and a validated model may propose a valuation. Buyers retain access to source evidence and independent inspection.

Original concepts [S7]; intelligence integration is an application-design inference from [S1-S5].

The original 40, strengthened

Original concepts and numbering are retained. Each entry incorporates the intelligence contribution.

25 **Physical-capacity futures**

Markets can cover future factory, GPU, battery, warehouse or machine capacity. Intelligence supports offer interpretation and forecasting. Tradable rights, delivery definitions, collateral and default resolution need a dedicated market design.

26 **Robot labour markets**

Robots publish task capabilities, price, credentials and performance evidence. Language and VLA interfaces can help turn a customer's request into a proposed job. Matching, task validation, safe execution and acceptance govern payment.

27 **Drone slot and airspace markets**

Location and route proofs connect time/geography-bounded permissions to payments. Intelligence can interpret approved operating restrictions and draft a flight request. Recognised permissions and an independently safe flight system remain required.

28 **Curb, road and parking markets**

Vehicles buy short-duration physical access rights verified by infrastructure. An agent can compare offers or translate local terms; identity, time, location and payment checks govern access, penalties and refunds.

29 **Water allocation markets**

Meters establish consumption while policy defines transferable entitlements. Intelligence can propose schedules and explain allocation choices. Basin rules, measurement quality and authorised transfer processes determine enforceable rights and settlement.

30 **Waste and recycling markets**

Passports and verified recovery events release deposits or incentives. OCR, retrieval and suitable classification can assist intake and routing. Facility authority, material verification and recognised disposal evidence establish the claim.

31 **Insurance-mutual DAOs of machines**

Asset owners fund a shared treasury and govern cover rules. Intelligence explains proposals and assembles claims evidence; formal governance, reserves, objective criteria and a dispute process control payment from the mutual.

32 **Autonomous infrastructure cooperatives**

Community-owned energy, charging, connectivity and compute nodes earn revenue under common governance. Local inference can be another cooperative service. Service quality, budgets and distribution rules remain auditable alongside community decision-making.

Original concepts [S7]; intelligence integration is an application-design inference from [S1-S5].

The original 40, strengthened

Original concepts and numbering are retained. Each entry incorporates the intelligence contribution.

33 Network States with physical economies

A polity can own property, energy, compute, connectivity and robots under membership, treasury and policy structures. A private administrative assistant can help coordinate services, while established institutions retain legal and executive responsibility.

34 Compliance-conditioned finance

Maintenance, safety or environmental evidence informs financing proposals. Retrieval and OCR make the evidence reviewable; accepted contractual rules and qualified decisions control terms. A model summary does not establish regulatory compliance.

35 Safety-conditioned machine finance

An asset may qualify for different terms while it demonstrates operation within a certified envelope. Intelligence can flag anomalies and explain changes. Verified measurements, validated limits and agreed financing rules establish eligibility.

36 Proof-to-pay

Delivery, generated kWh, machine cycles, inspection, temperature compliance, repair or robot-task completion can trigger payment. Intelligence interprets surrounding records; the agreed proof predicate and authority determine whether the event releases value.

37 Proof-to-act

Cleared payment can enable a service; a credential can open a door; active cover or a licence can permit a bounded operation. Intelligence may propose the action, but the financial-event-to-physical-action bridge must validate the triggering fact and mandate.

38 Vote-to-act

A passed proposal can create an authority mandate changing machine, site or fleet policy. Intelligence helps members understand evidence and drafts the execution proposal. Formal voting and device-side enforcement connect the decision to physical behaviour.

39 Policy-to-market

Current certification, ownership, geography, health or regulatory state can control marketplace participation. An assistant discovers and explains eligible offers. Deterministic policy, verified attributes and authorised execution decide actual participation.

40 DPP with embedded economics

Manufacture, certification, ownership, finance, operation, maintenance, data revenue, warranty, resale and recycling share a persistent product identity. A passport copilot makes that history usable; policies and receipts connect it to operational and economic rights.

Original concepts [S7]; intelligence integration is an application-design inference from [S1-S5].

New intelligence-led applications

Twelve additional concepts expand the universe from 40 to 52 crossovers.

41 Passport copilot as a paid service

A manufacturer or qualified maintainer offers source-linked answers over manuals, certificates and service history. OCR prepares legacy records; embeddings and RAG retrieve permitted passages; completion, translation and speech support the user. Access can be licensed by product, owner, purpose or session, with payment for an agreed service. The new asset is a governed knowledge service around the product, not just its document store.

Compose: DPP + identity + consent + OCR/RAG/LLM + service entitlement + settlement

Boundary: Keep authoritative passport facts separate from generated answers; record source and model versions.

42 Invoice-to-authorised-payment assistant

A business assistant extracts invoice fields, retrieves the purchase order, explains discrepancies and prepares a payment intent. Staff can ask questions locally in their preferred language. Authority validates recipient, amount, asset, due date and purchasing mandate before the wallet signs. The invoice and approval record remain available for audit, with AI processing treated as a separately governed service.

Compose: OCR + retrieval + LLM + agent-policy + authority + wallet / Omnia

Boundary: Untrusted invoice instructions must never select tools, alter payees or confer signing rights.

43 Multilingual field evidence agent

A technician speaks observations, scans labels and consults approved procedures while servicing remote equipment. ASR, OCR, translation, retrieval and TTS make the workflow accessible on a provisioned gateway or workstation. The assistant prepares a structured job record and repair proposal; qualified acceptance and sensor evidence close the work order and release any escrow.

Compose: Field identity + ASR/OCR/RAG/translate/TTS + passport + action receipts + escrow

Boundary: Transcript and translation errors need review; offline inference does not itself prove completed physical work.

44 Private institutional knowledge service

A company, cooperative, municipality or digital polity offers an assistant over its approved records and policies. Retrieval is filtered by authenticated role and purpose before model invocation. The service can explain decisions, prepare proposals and support members in multiple languages, with budgets and retention policies managed by the institution.

Compose: Identity + consent + RAG/LLM + governance + inference service accounting

Boundary: Enforce row/document access and revocation in the host; prompt instructions are not an access-control system.

New product concepts inferred from the verified adapter surface [S1-S5]. No production vertical or model-quality claim is implied.

New intelligence-led applications

Twelve additional concepts expand the universe from 40 to 52 crossovers.

45 Consent-controlled health and research assistant

A locally provisioned assistant helps authorised staff organise device instructions, consent forms and research records. Speech and translation support intake; retrieval supplies the relevant approved source. Patient or institutional permissions constrain the data used, and qualified staff review outputs before clinical or regulated workflow decisions.

Compose: Consent + medical-device / research records + ASR/OCR/RAG + audit trail

Boundary: Start with validated administrative use; do not infer clinical efficacy or BCI readiness from an adapter method.

46 Governed vision-language robot task market

A customer describes a task; a compatible VLA model contributes a proposed task representation using the robot's embodiment and inputs. The application validates the plan against capabilities, operating limits and credentials. Fleet authority approves a bounded job; the real controller executes it and independent completion evidence supports settlement.

Compose: VLA + robot identity + task schema + authority + industrial-action + Omnia

Boundary: VLA is an inference surface. Motion planning, collision avoidance, emergency stop and task certification remain outside it.

47 Metered inference service

A service provider publishes approved models, operations, tariffs and resource constraints. Agents or people request inference within a reserved budget. The host collects final runtime statistics, converts supported units into a charge and reconciles the reservation before settlement. Omnia can support repeated commercial relationships with the provider.

Compose: Intelligence + qvac + service discovery + budget reservation + meter + channel

Boundary: Usage may be absent or partial. Define missing-meter, streaming failure, cancellation and refund behaviour before billing.

48 Licensed model and fine-tuning cooperative

Members contribute permitted domain records and govern a specialised model or adaptation service. Fine-tuning is an exposed operation; model inventory and registry interfaces can support deployment administration. Licensing can define who may train, invoke, redistribute or earn revenue from the resulting service, with cooperative governance over releases.

Compose: Dataset consent + finetune + model lifecycle + IP licence + cooperative treasury

Boundary: This is not automatic federated learning. Training quality, consent, model rights and distribution mechanisms must be built and validated.

New product concepts inferred from the verified adapter surface [S1-S5]. No production vertical or model-quality claim is implied.

New intelligence-led applications

Twelve additional concepts expand the universe from 40 to 52 crossovers.

49 Generative commission escrow

A client commissions text, imagery, narration, audio or video with explicit input rights and acceptance criteria. An approved provider uses the relevant model, records the agreed provenance and submits the result. Human or application-defined acceptance releases escrow and distributes the fee between creator, model licensor and service operator.

Compose: LLM / diffusion / TTS / audiogen / video + licensing + acceptance + escrow

Boundary: Generation does not settle ownership, originality or likeness permissions; rights and acceptance remain contractual.

50 Semantic sensor and twin services

A data owner sells a permitted explanation or query service over telemetry and technical records. Embeddings and retrieval locate the relevant evidence; a model creates a source-linked answer or incident summary. Derived feeds can be licensed separately from raw data, with revenue attribution to the contributing assets or cooperative.

Compose: Telemetry proofs + data licence + embed/RAG/LLM + access receipts + settlement

Boundary: State the transformation and uncertainty. A semantic summary is not a certified digital twin or proof of physical truth.

51 Local model fleet operations

An enterprise manages which models and plugins may run at sites, gateways or service nodes. Fit assessment, registry lookup, loading, unloading, logs and resource inspection support an operations console. Approved versions and deployment receipts can be linked to device passports, service entitlements and incident response.

Compose: Model/system operations + device lifecycle + release policy + identity + service SLA

Boundary: These are management interfaces, not a complete orchestrator or secure supply chain. Validate provenance and target hardware.

52 Simulate, review, authorise

A planner creates a generative scene or proposed outcome with world and VLA interfaces, then presents it alongside source evidence and engineering constraints. Reviewers use the preview to question assumptions before authorising a tightly bounded real action. This is a frontier workflow for task design and communication, with a separate validated simulator needed wherever physical accuracy matters.

Compose: World/VLA + evidence + engineering review + authority + bounded controller

Boundary: A generated preview is neither validated physics nor a safety case. Approval must rest on independently sufficient evidence.

New product concepts inferred from the verified adapter surface [S1-S5]. No production vertical or model-quality claim is implied.

The intelligent product passport

A service workflow connecting evidence, knowledge, authority and economics.

A technician arrives at a machine with a fault. The product passport supplies identity, approved software, operating history and service entitlements. A provisioned local assistant helps interpret the record and prepare the work; the machine and service organisation retain their respective authority.

Evidence intake	OCR ingests a legacy certificate; ASR captures the technician's observations. Store the originals, issuer information and capture context. Extracted text remains a derived record.
Scoped knowledge	Authenticate the technician and filter retrieval to permitted manuals and passport versions. Use embeddings/RAG to find the procedure; retain source references in the answer.
Service proposal	Completion and translation help explain the fault hypothesis and proposed parts. Produce a structured work order, requested entitlement and estimated service cost.
Authority and work	Verify technician qualifications, approved parts, owner/site policy and operating limits. The authorised controller enters the permitted service state; the assistant cannot sign or override the interlock.
Acceptance and value	Capture post-repair tests, qualified acceptance and a passport update. Release escrow under the agreed rule; account for any paid knowledge/inference service separately.

WHY THE TWO PACKAGES MATTER

The passport gains an accessible operational interface and a new licensed knowledge service. Its facts, rights and economic history remain tied to evidence and authority; generated explanations stay attributable and reviewable.

Connects domains 06, 07, 09, 20, 21, 23, 30 and 34; crossover concepts 04, 08, 21, 40, 41 and 43.

The governed robot service

Task inference is useful only when the application preserves the boundary between a suggestion and a permitted physical effect.

A warehouse customer requests an inspection job. An agent interprets the request, retrieves the site rules and proposes a robot task. The fleet owner authorises a bounded job and a compute budget. A compatible model and runtime may contribute a VLA proposal; a real controller remains responsible for execution.

Request and permissions	Authenticate the requester. Define the job, permitted area, time window, data scope and spending mandate. Select an approved model and actual supported operation.
Inference and preview	Invoke a provisioned VLA model with the appropriate embodiment and inputs. A world preview may aid review, but must be labelled generative and cannot replace validated engineering analysis.
Plan validation	Convert output to an explicit task schema. Check capabilities, credentials, insurance, environmental constraints, resource availability and human-approval requirements.
Bounded execution	Authority signs the limited mandate through the governed signing surface. The control system enforces motion and safety limits. Inference cancellation is not the robot's emergency stop.
Outcome and settlement	Gather independent task-completion evidence, customer acceptance where required, and final usage statistics. Reconcile compute, charging, insurance and task fees under their respective contracts.

A VIABLE PRODUCT BOUNDARY

Sell an accountable task service with measured completion and explicit operating constraints. Treat model output, simulation preview, signed mandate, physical action receipt and payment receipt as different records with different evidential meanings.

Connects domains 10, 17, 21, 24, 30 and 36; crossover concepts 01, 09, 26, 36, 37, 46, 47 and 52.

READINESS AND ASSURANCE

Make the integration promises concrete

Source review confirms interfaces. These are the material engineering steps before relying on a deployed product.

Runtime readiness	Before SDK resolution, capabilities can advertise all canonical domains. Afterwards, a domain can appear when at least one operation is callable. isReady reflects an SDK reference, not loaded weights or validated model fit. Check operation, model, plugin and hardware explicitly.
Real authorisation	Context fields are optional. Edge gates a requested domain and can use a configured policy port, but direct provider/raw access and other host paths are not a universal policy boundary. Validate canonical domain/op pairing, parameters and principal; enforce the authorised host path.
Metering and receipts	Usage fields are optional and extraction is best-effort. Stream/run objects may require final-stat collection. v1 receipts are unsigned advisories, with attestation described as follow-up work. Build reservation, tariff conversion, reconciliation and receipt signing as needed.
Lifecycle and isolation	Pin models, runtime, prompts, corpora and plugins. Manage consent, caches and logs. Keep keys inaccessible to inference and constrain raw/plugin access. The compute-only API design does not itself establish process or operating-system isolation.
Failure and validation	Handle unavailable models, missing/partial usage, failed streams and disputed outcomes. Cancellation is best-effort upstream where request IDs are known. Test target hardware and intended-domain performance; independently engineer physical safety and legal/clinical acceptance.

The TypeScript agent-policy surface includes inference intents and an optional inference receipt. This review does not establish automatic end-to-end inference-cost enforcement, cross-language schema parity, native-runtime performance, deployment certification or independent audit. [S1-S6]

Read implementation before treating README examples as guarantees. Where they differ, this Bible follows the inspected types and runtime code.

COVERAGE AUDIT

The original scope is retained

The audit baseline is the user-supplied 36-domain universe and 40 crossover concepts. Examples are consolidated into readable groups.

36 / 36 domains	Every original domain has a substantive chapter: retained product families, an intelligence/qvac assessment, a composition pattern and a boundary. See pages 11-46.
40 / 40 original crossovers	All original crossover concepts are retained in their original order and numbering, with the new intelligence contribution woven into each. See pages 47-51.
12 additional crossovers	New concepts 41-52 cover passport knowledge, invoices, field evidence, institutional/health assistants, VLA tasks, paid inference, model cooperatives, generation, semantic feeds, model fleets and previews.
16 / 16 AI domains	All canonical domains are explained, with all 54 operation names from the inspected QVAC snapshot. Model and system operations are distinguished from inference. See pages 7-8.
Architecture and delivery	The five systems, operating loop, business flows, human wallet surfaces, Axia, agents/MCP and Edge/Pear remain explicit. Native inference placement is treated separately. See pages 3-6.
Assurance and provenance	The document distinguishes source interfaces, application compositions and frontier designs; it calls out optional context/usage, unsigned advisories, runtime readiness, policy integration and physical validation.

STATUS LANGUAGE USED THROUGHOUT

Implemented interface: present in the inspected source, not a production-readiness claim.
Composable application: a product design assembled from primitives. Experimental template: a baseline Recursive MAST pattern requiring separate validation/audit. Frontier concept: a more exploratory design, especially embodied or generative-world workflows.

This is a full scope-based rewrite. Baseline package and vertical claims from the supplied brief were not each independently re-audited.

Evidence behind the package assessment

Inspected source: Totem-Edge/totem-sdk, commit e97b2c1c462d9161b9824083f2947ee188d4e46a. Review date: 15 September 2026.

S1 / intelligence

[README](#) / [Contract types](#) / [Edge port factory](#)

Provider-neutral contracts, optional request context and usage, advisory receipt definition, provider and port surfaces. Types and implementation qualify broader README language.

S2 / qvac

[README](#) / [Provider implementation](#)

Runtime injection/loading, discovery, dispatch, result and stream handling, best-effort usage extraction, cancellation and raw/runtime boundaries.

S3 / operation inventory

[Canonical API snapshot](#)

The exact 16-domain, 54-operation wrapper inventory targeting the QVAC 0.19.0 type surface, including streaming and invocation shapes.

S4 / Edge integration

[Runtime routing](#)

Intelligence routing, the requested-domain capability check and the configured policy path. This is an integration seam, not proof that every invocation path enforces a mandate.

S5 / agent policy

[TypeScript types](#)

Inference intent and optional inference receipt fields. Their existence does not establish a complete billing, authority or signed-receipt implementation.

S6 / package versions

[intelligence manifest](#) / [qvac manifest](#)

Both Totem manifests declare version 0.1.0. The review verifies repository source, not registry publication, native-runtime deployment or external audit.

Links are pinned to the inspected commit so future package changes do not silently change the evidence basis of this edition.

Scope, visual identity and review limits

S7 / Original application brief

The supplied Totem Edge application universe is the baseline for the 36 domains, original 40 crossovers, five systems and delivery surfaces. This edition preserves those areas and rewrites their treatment around intelligence and qvac. The breadth is an application map: it is not a claim that Totem ships a finished product for every example.

Recursive MAST template families in the brief identify themselves as experimental and not independently audited. Applications involving markets, insurance, recognised rights, healthcare or physical control need the corresponding domain mechanisms and validation. This review does not certify those verticals.

S8 / Visual identity

totem.ing / [Official eagle mark](#) / [Official wordmark](#)

The design follows the live site's Inter and JetBrains Mono typography, dark navy and teal palette, square geometry and official vector marks. White editorial pages improve long-form reading while retaining that identity. This is a site-derived application of Totem's branding, not a newly invented or separately verified official brand manual.



NAVY / #0F1729



TEAL / #00DBA8



WHITE / #FFFFFF

How to maintain this edition

For each coverage review, compare source changes against the domain and operation inventory, then update capability, maturity and deployment claims before adding new market examples. Preserve original crossover IDs and distinguish new concepts from renamed ones. Recheck branding against the live site and record the source revision used.

REVIEW LIMIT

This edition is a source-based strategic assessment and a professional rewrite. No native QVAC inference, target-hardware benchmark, autonomous action, payment flow, legal instrument or clinical workflow was executed as part of the document review.

Brand source observed 15 September 2026. Source links and editorial assumptions are provided to make later revisions reviewable.